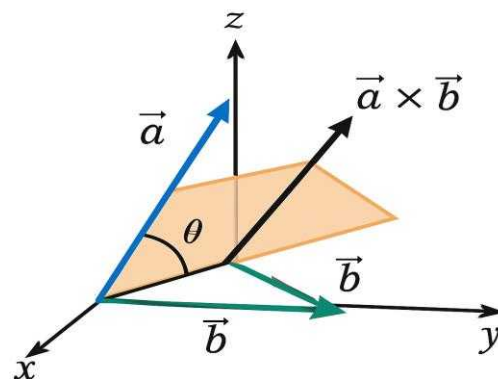


2. Cross Product (vector product)

The cross product is a mathematical operation between two vectors in 3D space. Unlike the dot product, which gives a scalar, the cross-product results in a new vector that is perpendicular to both input vectors and has a magnitude related to the angle between them.

$$\begin{array}{ccccc} & i & j & k & \\ & A_x & A_y & A_z & \\ & B_x & B_y & B_z & \\ A * B = & (A_y \cdot B_z - A_z \cdot B_y), & (A_z \cdot B_x - A_x \cdot B_z), & (A_x \cdot B_y - A_y \cdot B_x) & \end{array}$$

- $A = (1,0,0)$
- $B = (0,1,0)$
- $A*B = (0,0,1)$



2.1 Cross Product Applications

The cross product is a fundamental mathematical technique that has been used in many real-world applications across physics, engineering, and computer science. The cross product, as a vector used to produce a vector perpendicular to two others, makes it essential for understanding and calculating rotational effects, such as torque and angular momentum. In mechanical engineering, the cross product helps determine how forces cause objects to rotate, which is vital in designing engines, tools, and machines, as shown in Table No. 3.